PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM IV TAHOSSA NORTH WESTERN ZONE MOCK EXAMINATION 2025

013 GEOGRAPHY

TIME: 3HOURS AUGUST 2025

Instructions

1. This paper consists of sections A, B, and C, with eleven (11) questions.
2. Answer all questions in section A and B and only Two questions from section C
3. Section A carries sixteen (16) marks, Section B carries fifty-four (54) marks, and Section C carries thirty (30) marks.
4. A map extract of **KIGOMA (Sheet 92/3)** is provided.
5. Programmable calculators and cellular phones are not allowed in the examination room.
6. Write your examination on every page of your answer sheet provided.

SECTION A (16 Marks)
Answer ALL questions in this section

1) For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter in the answer booklet provided.

i) The government of Tanzania is constructing a hydroelectric power plant in the Rufiji River so as to ensure the availability of reliable electricity in the country. In what stage of the river is the river is the project taking place
   A. Middle stage
   B. D. Lower stage
   C. Upper stage
   D. Yazoo stage
   E. Delta stage

ii) The national census, which was conducted on August 2022, involved counting people who had stayed in the household during the night of enumeration. What kind of census was that?
   A. De jure
   B. Quinquennial
   C. Vital registration
   D. De facto
   E. Decennial

iii) When traveling in a fast-moving vehicle, we notice trees and other objects on both sides of the road moving in the opposite direction. This shows that the point of observation (the earth) is moving by rotation from:
   A. West to East
   B. E. West to South.
   C. East to West
   D. East to South
   E. West to North

iv) Dorine was explaining to his grandfather about layers of the earth's atmosphere. He told him that the atmosphere comprises with different layers in which one of them support rain formation due to its high amount of water vapour. Suggest that layer
   A. Mesosphere
   B. Atmosphere

C. Barysphere
D. Troposphere
E. Stratosphere

v) Which intrusive feature is a large, irregular-shaped body of igneous rock that cuts across existing rock layers?
A. Dike
B. Batholith
C. Sill
D. Laccolith
E. Phacolith

vi) What is the primary difference between the epicenter and the focus of an earthquake?
A. The epicenter is on the surface, the focus is underground
B. The epicenter is underground, the focus is on the surface
C. The epicenter and the focus are the same
D. The epicenter is the point of greatest damage
E. The focus is the point of greatest damage

vii) Imagine you are a geologist studying a landscape with a high concentration of limestone. Which chemical weathering process would you expect to be most prevalent and why?
A. Carbonation, because limestone reacts with carbonic acid
B. Solution, because limestone dissolves in water
C. Hydrolysis, because limestone reacts with water
D. Oxidation, because limestone contains iron
E. Hydration, because limestone absorbs water

viii) Imagine you are studying a rocky hillside. Over time, you observe that the rocks on the slope are slowly shifting downward due to the expansion and contraction caused by temperature variations. Which type of mass wasting involves this gradual, imperceptible movement?
A. Solifluction
B. Rock creep
C. Earthflow
D. Mudflow
E. Talus creep

ix) Suppose that you want to carry out a field study on the atmospheric conditions suitable for the growth of maize on a farm village. What are the best weather components to be considered?
A. Rainfall and wind
B. Rainfall and clouds cover
C. Moisture and sunshine
D. Rainfall and temperature
E. Pressure and humidity

x) Oceanographers have realized that water in the Indian Ocean contains a higher amount of salt than that of the Pacific Ocean. This means that salinity in ocean water increases with
A. Distance from the equatorial region
B. Depth of the sea water
C. Altitude
D. Increased rate of evaporation of the ocean
E. Increased rainfall

2) Match the description of the types of seismic waves in List A with their corresponding relevant answers in List B by writing a letter of the correct response beside the item Roman number in the answer sheet provided

LIST A	LIST B
i) These are the first to arrive at a seismic station, compress, and expand the ground like an accordion. ii) Waves move along the surface of the Earth, causing most of the shaking felt during an earthquake. iii) These waves move the ground up and down and side-to-side in the same direction that the wave is moving. iv) These waves cause the ground to move in an elliptical rolling motion, with particles moving in the same direction that the wave is moving. v) These waves travel through the interior of the Earth, including P-waves and S-waves. vi) These waves move the ground from side to side. Confined to the surface of the crust, they cause the shear or shake.	A. P-Waves B. Rayleigh Waves C. S-Waves D. Surface Waves E. Tsunami F. Love Waves G. Body Waves H. Ocean waves

SECTION B (54 Marks)

Answer all questions

3) (a) Briefly explain the types of ocean waves.
(b) Explain five factors affecting the rate of wave erosion.

4) Study the Map extract of KIGOMA (sheet 92/3), then answer the following questions:
   a) Calculate the area of Lake Tanganyika in $km^{2.}$
   b) Give the shape of the road at grid references 000584 and 960624.
   c) Find the distance of the coastal line in kilometers from grid reference 894560 to 970550.
   d) Why the coast of the north western part of the map is not highly populated?
   e) Find the bearing of the Nondwa point (893621) from Kitwe point (896556).

5) John was assigned to conduct research to determine the main source of noise pollution in Temeke. Help him to answer the following questions
   a) Suggest a suitable title for his research
   b) Give three sources of his research problem
   c) Briefly state three tools he used to collect data from respondents
   d) Name any two challenges he encountered when conducting this research

6) Imagine you are a geologist, last holiday you visited a unique landscape formed by chemical weathering processes in Tanga region. In the region, rainwater containing a weak carbonic acid reacted with the local rock, causing it to dissolve.
    a) What is the name of the landscape that you visited?
    b) Which process of chemical weathering was taking place in the area? Give a reason.
    c) By using a single diagram describe four holes that you observed on the area.

7) The Gamboshi district is planning to invite a survey team to help it determine the length of the Nyashimo to Ngasamo road.
    (a) What survey technique does the team decide to use?
    (b) Draw three (3) equipment the team can use in the survey.
    (c) List five (5) objectives of land survey

8) (a) Analyze the differences that exist between photographs and maps. 5points
    (b) Briefly explain the elements used in interpreting different photographs. 4 points

SECTION C (30 Marks)

Answer only **TWO** questions

9) The form four students of Maji ya Chai secondary school visited three different places and identified different types of soil with variation of physical and chemical properties. Explain six (6) physical properties of soil they observed.

10) According to census conducted in 2022 in our country, it has been observed that there is a great variation of the total population in our regions. Explain the environmental factors that have influenced the uneven population distribution in Tanzania.

11) With examples, explain any six ways used by a geographer to present statistical data